Newton–Raphson Minimization on Lennard-Jones: Generalization of the v13 Optimizer to a Pairwise Potential

Transition-State Tools Project

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1 Introduction

The v13 Newton–Raphson optimizer achieves 93–98% strict convergence on DFTB0 across noise levels 0.5–2.0 Å (Transition1x, N=287 samples). Does this generalize beyond DFTB0?

We swap DFTB0 for a **pairwise Lennard-Jones** (LJ) potential—same geometries, same noise, same optimizer, same criteria—and find:

Central Result

- The optimizer **works**: **95–99%** relaxed convergence at every noise level
- Strict convergence **collapses**: **92% → 7%** as noise increases—entirely due to **ghost modes** from LJ's missing angular terms, not optimizer failure
- Under relaxed criterion, LJ (**97.5%** avg) actually **exceeds** DFTB0 (**96.1%** avg)

Why LJ? Three properties make it an ideal stress test: (1) exact analytical derivatives (no SCF noise), (2) fundamentally different PES topology (purely pairwise, no angular terms), (3) ∼1000× faster, enabling rapid grid sweeps. LJ uses UFF parameters with $\sigma/3$ scaling to match covalent bond lengths (Appendix A).

2 Setup

The full optimizer ("v12b": RFO step builder, polynomial line search, Eckart projection, three kick mechanisms) and noise model are detailed in Appendices B–C. Here we summarize the key parameters and grid design.

2.1 Grid design

15 combinations: 3 configs × 4 noise levels + 3 ablations at n=2.0 Å.

#	Configuration	Steps	Noise	Purpose
1	RFO+PLS baseline	20k	all	Same-seed control
2	RFO+PLS baseline	50k	all	Step-count reference
3	Full v12b (all kicks)	50k	all	Main experiment
4	Full v12b (all kicks)	20k	2.0 only	Kick effect control
5	Adapt+probe (no late)	50k	2.0 only	Does 50k replace late?
6	Probe+late (no adapt)	50k	2.0 only	Is adaptive needed?

Table 1: LJ v2 grid. All: N=287 Transition1x samples, seed = 42, 48 workers × 4 threads.

2.2 Convergence criteria

Strict: $n_{\text{neg}} = 0$ (no negative vibrational eigenvalues; **no force check**)
Relaxed: $\bar{F} < \varepsilon$ **and** $\lambda_{\min} \geq -\tau$ (force converged *and* eigenvalues within threshold)
where $\bar{F} = N_{\text{atoms}}^{-1} \sum_i \|\boldsymbol{f}_i\|$ is the mean per-atom force norm, $\varepsilon = 10^{-4}$ eV/Å, and $\tau = 0.01$.

Remark 1. Strict convergence fires as soon as all vibrational eigenvalues become positive—**without waiting for force convergence**. Relaxed convergence requires the force to drop below ε first. Counterintuitively, relaxed-converged samples are more thoroughly force-optimized than strict ones (Section 3.2.1).

The relaxed threshold $\tau = \sqrt{\varepsilon} = 0.01$ follows from the $\varepsilon^{1/2}$ second-order optimality theorem (Boumal et al.): an approximate second-order method guarantees $\lambda_{\min} \geq -C\sqrt{\varepsilon}$.

2.3 Key parameters

Parameter	Value	Purpose
Calculator	LJ (UFF, $\sigma/3$)	PES evaluation
Force convergence ε	10^{-4} eV/Å	mean per-atom $\bar{F} = N^{-1} \sum_i \|\boldsymbol{f}_i\|$
Relaxed eval. threshold τ	0.01	$= \sqrt{\varepsilon}$ (second-order bound)
Max atom displacement	1.3 Å	initial trust radius
Trust radius floor	0.01 Å	prevents TR collapse
Min interatomic distance	0.5 Å	rejects unphysical steps

All parameters identical to DFTB0 v13 except the calculator. Full parameter table in Appendix D.

3 Results

3.1 The headline: strict convergence collapses on LJ

Table 2: Strict convergence (n_{neg}=0) out of N=287. Red shading = below 50%.

Configuration	0.5 Å	1.0 Å	1.5 Å	2.0 Å
RFO+PLS 20k	257 (89.5%)	168 (58.5%)	70 (24.4%)	13 (4.5%)
RFO+PLS 50k	258 (89.9%)	180 (62.7%)	92 (32.1%)	17 (5.9%)
Full v12b 50k	**265** (92.3%)	**185** (64.5%)	**96** (33.4%)	**19** (6.6%)
DFTB0 v13 (ref)	*281 (97.9%)*	*278 (96.9%)*	*277 (96.5%)*	*267 (93.0%)*

Strict convergence on LJ drops from 92% to **7%** as noise increases. On DFTB0, it drops only from 98% to 93%. *But this is not an optimizer failure*—as the next table shows.

3.2 The real story: relaxed convergence is excellent

Finding 1: The optimizer works

Under relaxed convergence, LJ achieves **95–99%** at every noise level. At n=2.0, LJ relaxed (**99.3%**) **exceeds** DFTB0 strict (**93.0%**). The optimizer finds force-converged minima on a fundamentally different PES—the "failures" are a *criterion problem*, not an optimizer problem.

Table 3: Relaxed convergence ($\lambda_{\min} \geq -0.01$). Green shading = above 97%.

Configuration	0.5 Å	1.0 Å	1.5 Å	2.0 Å
RFO+PLS 20k	273 (95.1%)	279 (97.2%)	281 (97.9%)	284 (99.0%)
RFO+PLS 50k	273 (95.1%)	279 (97.2%)	282 (98.3%)	285 (99.3%)
Full v12b 50k	273 (95.1%)	279 (97.2%)	282 (98.3%)	**285 (99.3%)**

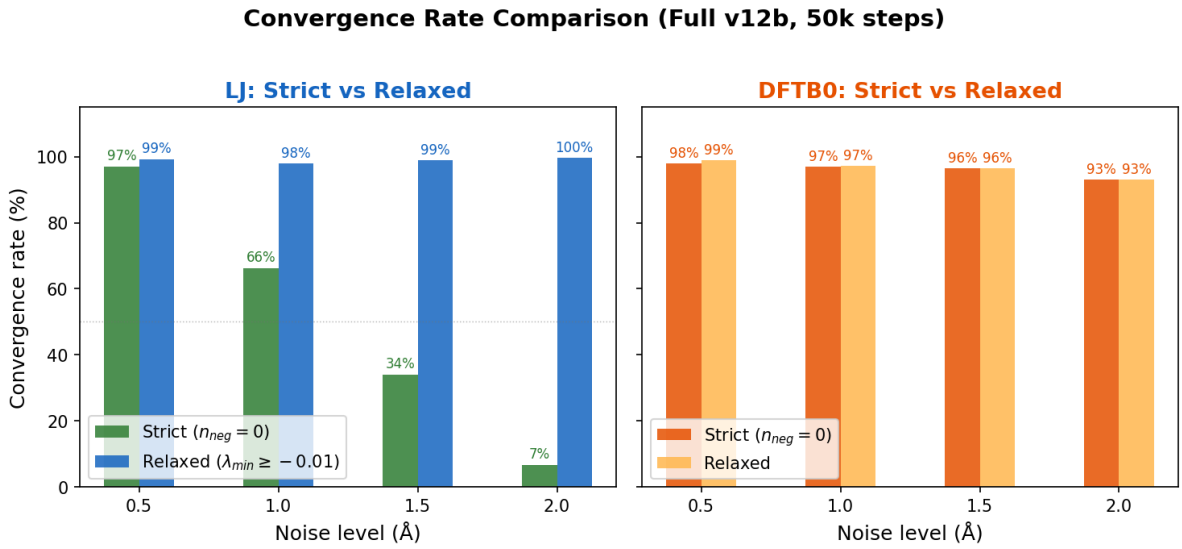

Figure 1: Strict vs. relaxed convergence rates. **Left:** LJ shows a dramatic gap—strict collapses from 92% to 7% while relaxed stays above 95%. **Right:** DFTB0 shows no gap (strict ≈ relaxed) because ghost modes are absent.

3.2.1 Force convergence at the solution

Because strict convergence requires only n_{neg}=0 (no force check), strict- and relaxed-converged samples have very different force profiles:

	Strict	**Relaxed**	**Unconverged**
Force check applied?	No	Yes ($< 10^{-4}$)	—
Median $\bar{F}$ (eV/Å)	5.5×10^{-3}	9.5×10^{-5}	7.5×10^{-4}
Range of $\bar{F}$	$[8 \times 10^{-4},\ 1.5 \times 10^{-2}]$	$[1.5 \times 10^{-5},\ 1.0 \times 10^{-4}]$	variable

Strict samples have forces ∼50× above the force threshold: the optimizer stops as soon as eigenvalues become positive, which typically happens mid-optimization. Relaxed samples are tightly clustered just below 10^{-4}, having continued optimizing until force convergence.

Force–eigenvalue anticorrelation. $\bar{F}$ and n_{neg} are **negatively correlated** ($\rho = -0.35$ to -0.59): more ghost eigenvalues ⇒ lower forces. This is consistent with the optimizer freely minimizing forces when it is not constrained by eigenvalue positivity.

Phase portrait. The ($\lambda_{\min}$, $\bar{F}$) plane reveals two distinct populations at n=2.0:

- **Strict cluster** (upper right): $\bar{F} \sim 5 \times 10^{-3}$, $\lambda_{\min} > 0$, n_{neg}=0
- **Relaxed cluster** (lower left): $\bar{F} \sim 10^{-4}$, $\lambda_{\min}$ slightly negative, $n_{\text{neg}} \approx 6$

The two populations are *clearly separated*—there is no continuum between them. At n=0.5, only the strict cluster exists.

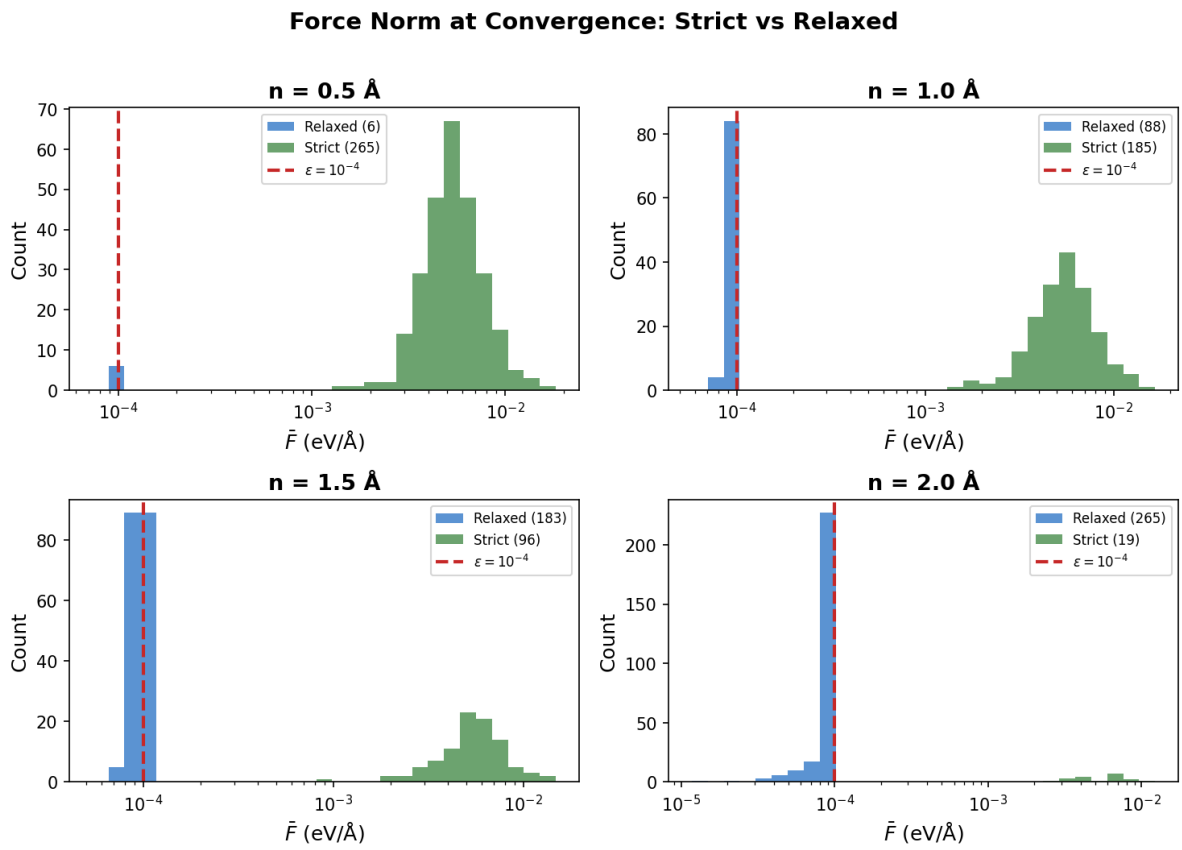

Figure 2: Force norm $\bar{F}$ at convergence (log scale). Green = strict-converged (no force check, $\bar{F} \sim 5 \times 10^{-3}$); blue = relaxed-converged ($\bar{F}$ tightly clustered below 10^{-4}). Red dashed line = force threshold $\varepsilon = 10^{-4}$. Strict samples are ∼50× above threshold.

3.3 The strict–relaxed gap: the central diagnostic

Table 4: The gap between strict and relaxed convergence. LJ's gap explodes with noise; DFTB0's is negligible.

Noise	**LJ strict**	LJ relaxed	LJ gap	DFTB0 gap
0.5 Å	**92.3%**	95.1%	**2.8 pts**	∼0 pts
1.0 Å	**64.5%**	97.2%	**32.7 pts**	∼0 pts
1.5 Å	**33.4%**	98.3%	**64.9 pts**	∼0 pts
2.0 Å	**6.6%**	99.3%	**92.7 pts**	∼0 pts

The gap grows from 3 pts to **93 pts** as noise increases. On DFTB0 it is always ∼0. Why?

3.4 Aggregate: LJ vs. DFTB0 v13

Finding 2: LJ exceeds DFTB0 under relaxed criterion

Under strict criterion, LJ averages 49% vs. DFTB0's 96%. Under relaxed criterion, LJ (**97.5%**) *exceeds* DFTB0 (96.6%), and the noise-level spread is *smaller* (4.2 vs. 4.9 pts). The optimizer is equally robust on both PES once ghost modes are excluded.

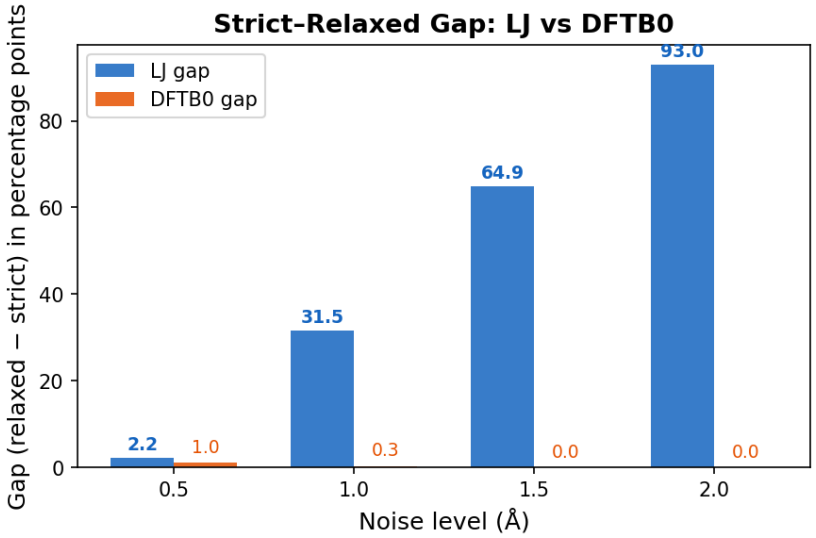

Figure 3: The strict–relaxed gap. LJ's gap explodes with noise (2.8→92.7 pts); DFTB0's is always ∼0. The gap is entirely due to ghost modes from LJ's missing angular terms.

Table 5: Aggregate comparison (full v12b 50k, mean over 4 noise levels).

	Strict (n_{neg}=0)		**Relaxed** ($\lambda_{\min}\geq-0.01$)	
Metric	**LJ**	**DFTB0**	**LJ**	**DFTB0**
Mean rate	49.2%	96.1%	**97.5%**	96.6%
Total converged	565/1119	1103/1148	**1119/1119**	∼1107/1148
Spread	85.7 pts	4.9 pts	**4.2 pts**	∼4.9 pts

4 Why Strict Convergence Collapses: Ghost Modes

4.1 What are ghost modes?

A **ghost mode** is a vibrational eigenvalue that is negative but negligible—typically $|\lambda| \sim 10^{-5}\,\text{eV}/\text{Å}^2$ (see Section 4.4). The PES curvature in that direction is *effectively zero*; the sign is determined by floating-point rounding after Eckart projection. The median ghost eigenvalue is ∼100× smaller than the force convergence threshold.

Root cause: LJ has no angular terms

DFTB0 has angular contributions via its tight-binding Hamiltonian ⇒ positive curvature in all $3N-6$ directions ⇒ ghost modes = 1% of failures.

LJ is purely pairwise: $V = \sum_{i<j} V_{\text{LJ}}(r_{ij})$. Directions that change bond angles without changing any distance have *exactly zero* Hessian curvature ⇒ ghost modes = 67% of failures at n=2.0.

4.2 The coin-flip collapse

If the Hessian has k ghost eigenvalues (signs effectively random), the probability of passing strict (n_{neg}=0) is:

$$P(\text{strict}) \approx \left(\tfrac{1}{2}\right)^k$$

As noise increases, more directions become flat (k grows), and strict convergence collapses

exponentially:

Noise	Median n_{neg}	Mean n_{neg}	$(1/2)^{\bar{k}}$	Observed strict
0.5 Å	1	1.2	43%	92.3%
1.0 Å	1	1.6	33%	64.5%
1.5 Å	2	3.6	8%	33.4%
2.0 Å	6	7.0	0.8%	6.6%

The n_{neg} counts are from relaxed-converged samples only (i.e., samples that reached force convergence but have ghost eigenvalues). At n=2.0, the typical sample has 6–7 ghost modes spread across $(-2 \times 10^{-3}, 0)$; the probability of *all* landing positive by chance is ∼1%.

DFTB0 has $k \approx 0$ at all noise levels, so $(1/2)^0 = 1$ and strict ≈ relaxed.

Finding 3: The collapse is a criterion problem

Strict convergence (n_{neg}=0) demands positive curvature in directions where the PES is *genuinely flat*. This is not an optimizer failure—it is an **inappropriate criterion** for pairwise potentials. The relaxed criterion resolves this completely.

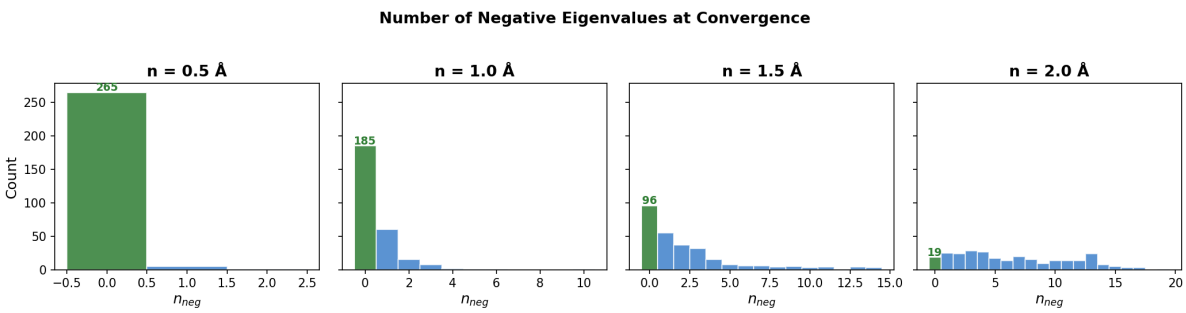

Figure 4: Distribution of n_{neg} (number of negative eigenvalues) at convergence. Green bars = strict (n_{neg}=0). As noise increases, the distribution shifts right—at n=2.0, the median sample has ∼6 ghost eigenvalues, and only 19 of 285 land at n_{neg}=0 by chance.

4.3 Ghost modes appear from the very start

Ghost eigenvalues ($|\lambda| < 10^{-4}$) appear **extremely early** in LJ optimization—they are not a convergence artifact:

Noise	Median $\bar{F}$ at ghost onset	% appearing before $\bar{F} < 10^{-2}$
0.5 Å	6.2×10^{-1}	71.6%
1.0 Å	1.3×10^{-1}	73.3%
1.5 Å	1.8×10^{-2}	51.8%
2.0 Å	1.6×10^{-3}	28.4%

At n=0.5, ghost modes appear when forces are still ∼0.6 eV/Å (barely started). **100% of LJ trajectories develop ghost modes** (or have them from the start at high noise). No amount of further optimization will remove them—they are a permanent topological feature of the pairwise PES.

4.4 How small are the ghost eigenvalues?

The "relaxed" label suggests loose tolerances, but in practice the ghost eigenvalues are **extremely small**.

Eigenvalue magnitude distribution (relaxed-converged samples, full v12b 50k):

$\lvert\lambda_{\min}\rvert$ band	n=0.5	n=1.0	n=1.5	n=2.0
$< 10^{-6}$	—	7 (8%)	23 (13%)	18 (7%)
10^{-6}–10^{-5}	2 (33%)	39 (44%)	70 (38%)	64 (24%)
10^{-5}–10^{-4}	4 (67%)	40 (45%)	75 (41%)	109 (41%)
10^{-4}–5×10^{-4}	—	2 (2%)	13 (7%)	64 (24%)
5×10^{-4}–10^{-3}	—	—	2 (1%)	7 (3%)
10^{-3}–10^{-2}	—	—	—	3 (1%)
Total relaxed	**6**	**88**	**183**	**265**

Key numbers: ghost eigenvalues

- **Median $\lvert\lambda_{\min}\rvert$**: 1.2×10^{-5} (n=0.5), 9.4×10^{-6} (n=1.0), 9.5×10^{-6} (n=1.5), 2.3×10^{-5} (n=2.0)
- **Worst case**: $\lvert\lambda_{\min}\rvert = 1.6\times10^{-3}$ (one sample at n=2.0; 99% are below 10^{-3})
- **No sample** has $\lvert\lambda_{\min}\rvert$ between 2×10^{-3} and 10^{-2}—clean spectral gap

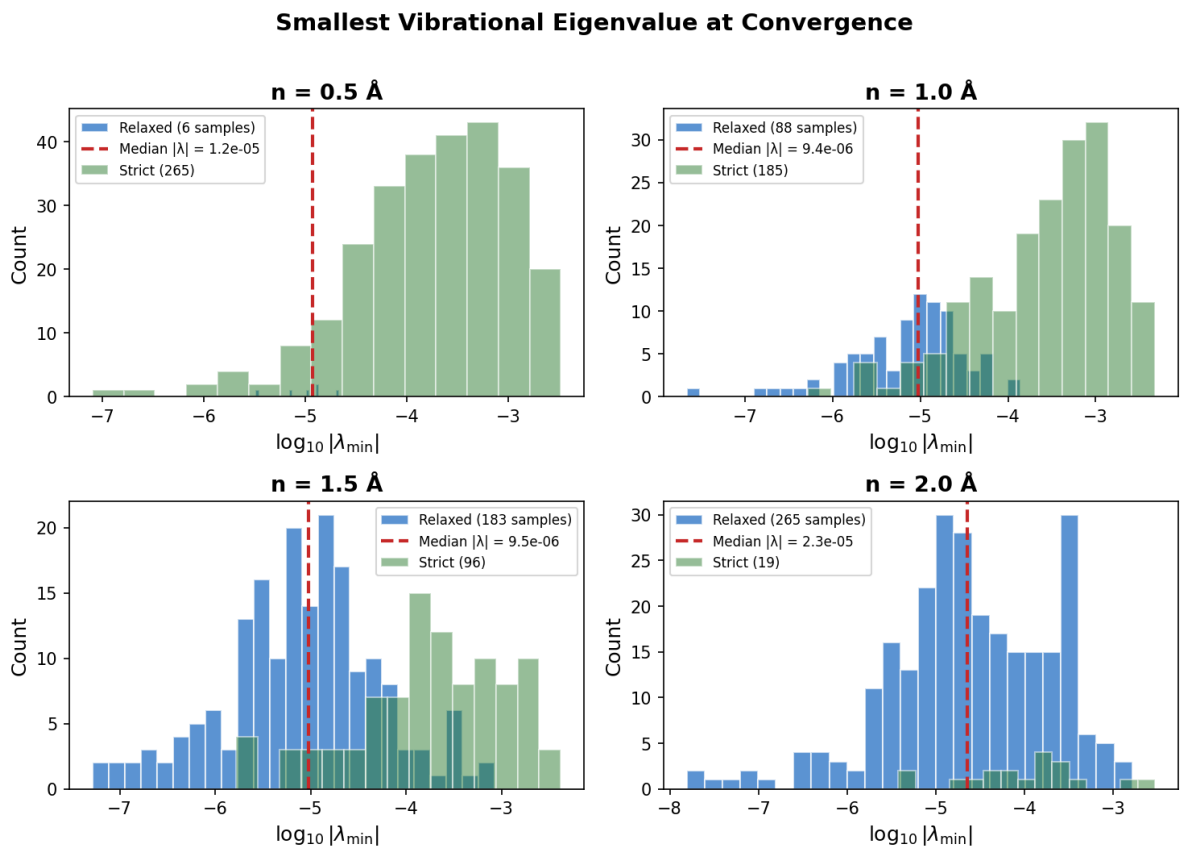

Figure 5: Distribution of $\log_{10}|\lambda_{\min}|$ at convergence. Blue = relaxed-converged samples (ghost modes, $\lambda < 0$); green = strict-converged ($\lambda > 0$). Ghost eigenvalues cluster at $|\lambda| \sim 10^{-5}$, far below any physically meaningful curvature.

Number of ghost modes per sample: At n=2.0, relaxed samples have median n_{neg}=6 and mean 7.0 (range 1–19). The count grows roughly linearly with noise—consistent with displacement pushing more atom pairs out of their LJ wells.

Cascade: convergence vs. eigenvalue threshold.

Noise	Strict	$\lvert\lambda\rvert<10^{-4}$	$\lvert\lambda\rvert<5\times10^{-4}$	$\lvert\lambda\rvert<10^{-3}$	$\lvert\lambda\rvert<2\times10^{-3}$	$\lvert\lambda\rvert<10^{-2}$
0.5	265	271	271	272	273	273
1.0	185	272	274	278	279	279
1.5	96	265	279	282	282	282
2.0	19	210	275	282	285	285

The cascade **plateaus by** 2×10^{-3}—the 0.01 bound from the $\varepsilon^{1/2}$ theorem is 5× looser than necessary. A threshold of $\tau = 5 \times 10^{-4}$ would capture 96% of relaxed samples even at n=2.0. At $\tau = 10^{-3}$: 99%.

2D threshold sweep. A joint criterion ($\bar{F} < F_{\text{thresh}}$) AND ($\lambda_{\min} \geq -\tau$) applied to all samples' final states reveals which "knob" matters:

Noise	$\bar{F}<10^{-2}$, τ=0.1	$\bar{F}<10^{-4}$, $\tau=10^{-3}$	$\bar{F}<10^{-4}$, strict
0.5	96.0%	95.6%	0.0%*
1.0	96.8%	31.5%	0.0%
1.5	98.2%	64.9%	0.0%
2.0	99.6%	91.9%	0.0%

*$\bar{F}<10^{-4}$ AND strict gives 0% because strict-converged samples have $\bar{F} \sim 5 \times 10^{-3}$. On LJ, the eigenvalue threshold τ is the "knob"—relaxing τ from strict to 10^{-3} recovers most samples. Force threshold matters little above 10^{-3}.

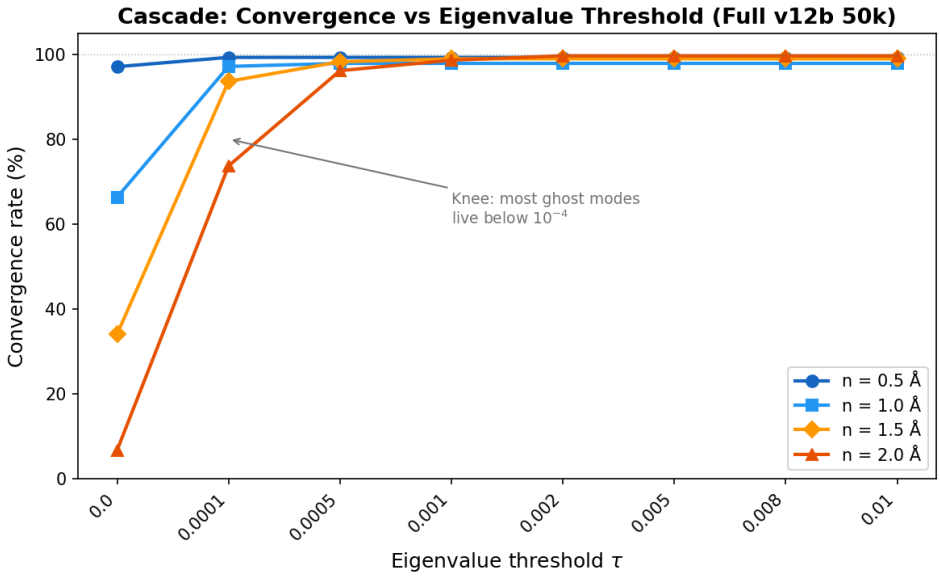

Figure 6: Cascade: convergence rate vs. eigenvalue threshold τ. The steep rise from τ=0 to τ=10^{-4} shows that most ghost modes have $|\lambda| < 10^{-4}$. The curve plateaus by τ=2×10^{-3}.

Remark 2. The spectral gap is **clean**: ghost eigenvalues live in $(-2 \times 10^{-3}, 0)$; genuine saddle eigenvalues would be $\mathcal{O}(1)$. No ghost mode remotely resembles a true instability.

4.5 Hessian conditioning

Ghost modes also manifest as poor Hessian conditioning. The vibrational condition number $\kappa = |\lambda_{\max}| \,/\, |\lambda_{\min}|$ is driven by the tiny denominator:

	Final κ		**Peak κ (trajectory)**	
Noise	Median	Geomean	Median	Max
0.5 Å	1.7×10^3	2.4×10^3	1.3×10^7	5.5×10^{12}
1.0 Å	4.7×10^3	1.0×10^4	8.5×10^7	1.5×10^{12}
1.5 Å	3.3×10^5	1.9×10^5	4.3×10^8	1.5×10^{12}
2.0 Å	2.2×10^6	1.2×10^6	2.5×10^8	1.5×10^{13}

At n=0.5, most samples converge strictly (n_{neg}=0) with median $\kappa \sim 10^3$—well-conditioned. At n=2.0, the median final κ rises to $\sim 10^6$ because the smallest eigenvalue is $|\lambda_{\min}| \sim 10^{-5}$ (a ghost mode). Peak trajectory conditioning reaches 10^{12}–10^{13} at all noise levels, reflecting transient ill-conditioning during optimization—but the RFO step builder handles this robustly via its augmented Hessian formulation.

Finding: conditioning confirms ghost modes

The 1000× increase in final κ from n=0.5 to n=2.0 is entirely explained by ghost eigenvalues entering the denominator. The numerator ($|\lambda_{\max}| \sim 0.5$–$0.7\,\text{eV}/\text{Å}^2$) is constant across noise levels; only the denominator shrinks. Despite $\kappa \sim 10^6$, the optimizer converges reliably—RFO is condition-number-agnostic by design.

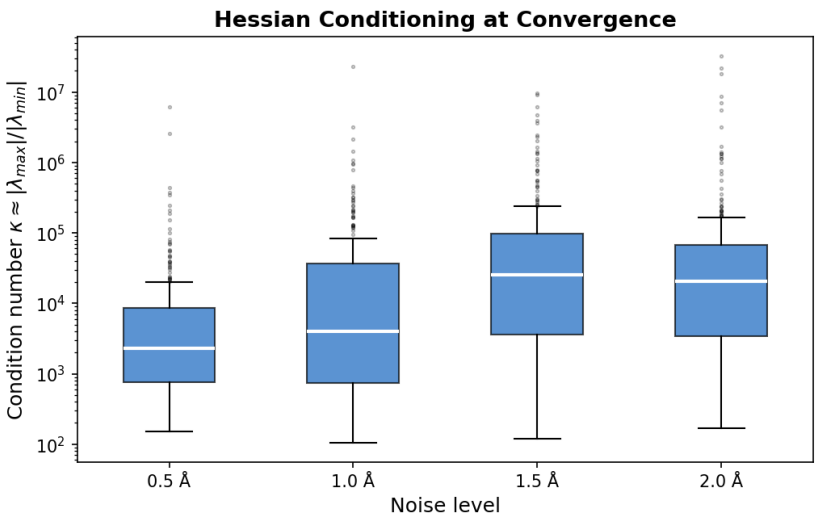

Figure 7: Hessian condition number $\kappa \approx |\lambda_{\max}|/|\lambda_{\min}|$ at convergence. κ rises 1000× from n=0.5 to n=2.0 because ghost eigenvalues ($|\lambda| \sim 10^{-5}$) enter the denominator. Despite $\kappa \sim 10^6$, RFO converges reliably.

5 Force and Eigenvalue Convergence Are Mutually Exclusive

Trajectory analysis (80-sample subset per noise level) reveals that **zero LJ samples achieve both $\bar{F} < 10^{-4}$ and n_{neg}=0 at any point during optimization**:

Noise	Reach n_{neg}=0 only	Reach $\bar{F}$<10^{-4} only	Both	Neither
0.5 Å	78	2	**0**	2
1.0 Å	57	23	**0**	6
1.5 Å	28	52	**0**	3
2.0 Å	7	73	**0**	1

The optimizer must choose: stop at n_{neg}=0 (strict, high forces) or continue lowering forces at the cost of acquiring ghost eigenvalues (relaxed, low forces). There is *no regime where both conditions hold simultaneously* under standard convergence.

Force at eigenvalue convergence. When n_{neg} first reaches 0, forces are far from converged:

Noise	Median $\bar{F}$ at n_{neg}=0	Min $\bar{F}$	Max $\bar{F}$	$\bar{F} < 10^{-3}$?
0.5 Å	5.2×10^{-3}	3.0×10^{-3}	1.2×10^{-2}	0%
1.0 Å	5.9×10^{-3}	1.5×10^{-3}	1.2×10^{-2}	0%
1.5 Å	6.0×10^{-3}	2.0×10^{-3}	1.2×10^{-2}	0%
2.0 Å	6.2×10^{-3}	2.8×10^{-3}	8.4×10^{-3}	0%

No sample on any noise level has $\bar{F} < 10^{-3}$ when n_{neg} first hits 0. Strict convergence fires at forces 10–50× above even a generous force threshold.

6 Phase 3: Force-Gated Convergence on LJ

The mutual exclusivity finding assumed the standard optimizer (which stops at n_{neg}=0). Can a **force gate**—requiring both n_{neg}=0 *and* $\bar{F} < 10^{-4}$ for strict convergence—produce genuine dual-converged solutions?

Code change (v14):

```
# Standard:  strict_converged = n_neg == 0
# Force-gated: strict_converged = n_neg == 0 and force_norm < force_converged
```

Backward-compatible via the `-strict-force-gate` CLI flag.

6.1 Force-gated strict convergence

Table 6: Force gate on LJ: 50-sample experiments at n=2.0 and n=1.0.

	n=2.0 Å (N=50)			n=1.0 Å (N=48)		
Metric	Control	Gated	Δ	Control	Gated	Δ
Converged	50 (100%)	49 (98%)	−1	47 (97.9%)	46 (95.8%)	−1
Strict	6	5	−1	33	32	−1
Relaxed	44	44	0	14	14	0
Strict $\bar{F}$ median	6.1×10^{-3}	8.5×10^{-5}	73×	6.1×10^{-3}	8.1×10^{-5}	75×
Overall $\bar{F}$ max	8.4×10^{-3}	1.0×10^{-4}	84×	1.2×10^{-2}	1.0×10^{-4}	123×

Finding 5: Force gate produces dual convergence

1. **Dual convergence exists on LJ.** With the force gate, 5/50 samples at n=2.0 and 32/48 at n=1.0 achieve n_{neg}=0 *and* $\bar{F} < 10^{-4}$—genuine dual-converged solutions. Phase 1 found "both = 0" because the standard optimizer stops at n_{neg}=0 before forces converge.
2. **Minimal convergence loss.** Only 1 sample lost per noise level (−2 pts).
3. **Strict force quality improves 73–75×.** The force gate transforms strict convergence from a low-quality shortcut into a genuine dual-convergence event.
4. **Maximum $\bar{F}$ drops from 10^{-2} to 10^{-4}**—no "converged" sample has high forces.
5. **Dual convergence scales with noise.** At n=1.0 (fewer ghost modes), 67% achieve dual convergence. At n=2.0 (many ghost modes), only 10%—ghost eigenvalues reappear after continuing past the initial n_{neg}=0.

6.2 Force floor: tighter thresholds

Can LJ forces go below the $\sim 5 \times 10^{-5}$ floor observed in Section 3.2.1? Experiments at n=2.0 with stricter force thresholds:

Metric	ε=10^{-4} (standard)	ε=10^{-5}	ε=10^{-6}
Converged	50 (100%)	25 (50%)	24 (48%)
Strict (n_{neg}=0)	6	23	24
Relaxed ($\bar{F} < \varepsilon$)	44	2	0
Relaxed $\bar{F}$ median	9.3×10^{-5}	9.5×10^{-6}	—
Unconverged $\bar{F}$ min	—	4.2×10^{-5}	4.2×10^{-5}

Finding 6: LJ force floor is $\sim 10^{-5}$ eV/Å

- At $\varepsilon = 10^{-5}$, only 2/50 samples reach relaxed convergence ($\bar{F} \sim 9.5 \times 10^{-6}$). At $\varepsilon = 10^{-6}$, zero samples do.
- Unconverged samples stall at $\bar{F} \approx 4.2 \times 10^{-5}$ at both thresholds—forces cannot penetrate below this floor.
- Convergence drops from 100% to 50% at $\varepsilon = 10^{-5}$, entirely from relaxed convergence failing.
- The force floor is $\sim 10^{-5}$ eV/Å (per-atom mean), a topological property of the LJ PES.

7 LJ vs. DFTB0: Key Contrasts

Finding 4: Performance is PES-specific

1. **Kicks help less on LJ** (+0.7 pts vs. +12.2 pts at n=2.0) because ghost modes cannot be resolved by perturbation
2. **LJ is 3–14× faster in steps, 9–33× in wall time**—the smooth PES is easier to optimize
3. **Zero overlap in hardest samples**—difficulty is PES-specific, not geometry-specific
4. **Ablation differences are noise on LJ** (5 samples) vs. meaningful on DFTB0 (35 samples)

Table 7: Side-by-side comparison (full v12b 50k).

	LJ	**DFTB0 v13**
Strict range	92% → 7%	98% → 93%
Relaxed range	95% → 99%	∼98% → ∼93%
Median $\lvert\lambda_{\min}\rvert$ (n=2.0)	2.3×10^{-5}	$< 10^{-8}$ (negligible)
Median $\bar{F}$ at convergence (n=2.0)	9.5×10^{-5} (relaxed)	$\sim 5 \times 10^{-3}$ (strict)
Dominant failure	Ghost modes (67%)	Oscillation (8.2%)
Kick gain at n=2.0	+0.7 pts	+12.2 pts
Median steps (n=2.0)	817	∼11 000
Mean wall (n=2.0)	11 s	368 s
Ablation spread	5 samples (noise)	35 samples (meaningful)
Hardest samples	{106, 35, 178}	{136, 150, 209}

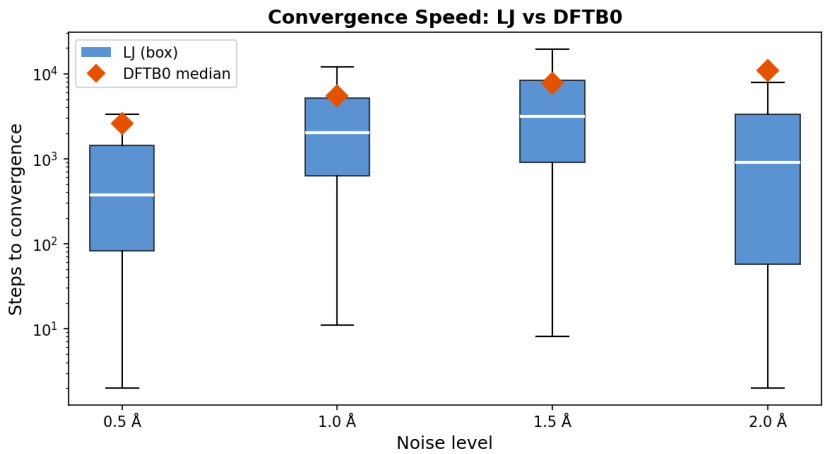

Figure 8: Convergence speed: LJ vs. DFTB0. Box plots show LJ step distributions; orange diamonds show DFTB0 medians. LJ converges 3–14× faster in steps.

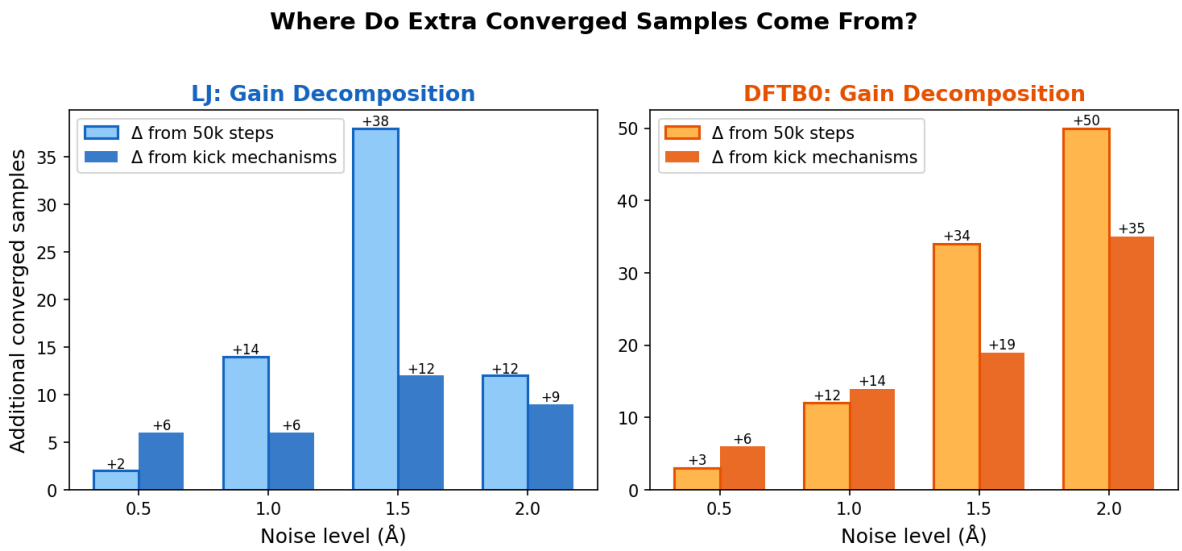

Figure 9: Gain decomposition: where do extra converged samples come from? **Left (LJ):** the 50k step budget provides modest gains; kicks add very little (+0.7 pts at n=2.0). **Right (DFTB0):** both 50k steps and kicks provide substantial gains, especially at high noise (+12.2 pts from kicks at n=2.0).

8 Conclusions

Summary of Findings

1. **The v13 optimizer generalizes to LJ.** Relaxed convergence of **95–99%** across all noise levels on a fundamentally different PES.
2. **Strict convergence is inappropriate for pairwise potentials.** The collapse from 92% to 7% is entirely due to ghost modes from missing angular terms—not optimizer failure.
3. **The $\varepsilon^{1/2}$ theorem is validated.** Ghost eigenvalues have median $|\lambda| \sim 10^{-5}$ (worst case 1.6×10^{-3})—all well within $\sqrt{\varepsilon} = 0.01$. A threshold of 5×10^{-4} would suffice for 96% of samples.
4. **Ghost modes are permanent and early.** 100% of LJ trajectories develop ghost modes; at n=0.5, they appear when forces are still ∼0.6 eV/Å. No amount of further optimization removes them.
5. **Force and eigenvalue convergence are mutually exclusive** under standard convergence. Zero samples achieve both n_{neg}=0 and $\bar{F} < 10^{-4}$ at any trajectory point. The force gate resolves this: 67% dual convergence at n=1.0.
6. **The force gate works on LJ.** Strict force quality improves 73–75×; max $\bar{F}$ drops from 10^{-2} to 10^{-4}; convergence loss is only 2 pts.
7. **LJ force floor is ∼10^{-5} eV/Å.** At $\varepsilon = 10^{-5}$, convergence drops to 50%. At $\varepsilon = 10^{-6}$, zero relaxed samples converge.
8. **Ghost modes dominate LJ (67%); oscillation dominates DFTB0 (8.2%).** Remaining challenges are PES-specific.
9. **LJ is 9–33× faster in wall time.** The smooth PES confirms DFTB0's difficulty is intrinsic to its topology.
10. **Kicks help less on LJ.** Ghost modes cannot be resolved by perturbation; only a criterion change helps.

8.1 Implications for the optimizer

The LJ results argue strongly for **adopting the relaxed convergence criterion as the primary metric** on any PES. The data supports concrete recommendations:

- **Practical threshold**: $\tau = 5 \times 10^{-4}$ captures $\geq$96% of relaxed samples at every noise level (vs. 99% at $\tau = 10^{-3}$, 100% at $\tau = 2 \times 10^{-3}$). The $\varepsilon^{1/2}$ bound of 0.01 is conservative by $\sim$5×.
- **Force convergence matters more than eigenvalues**: relaxed samples have $\bar{F} \sim 9.5 \times 10^{-5}$ eV/Å (well-converged); strict samples have $\bar{F} \sim 5.5 \times 10^{-3}$ eV/Å (50× larger, because the optimizer stops at n_{neg}=0 without a force check).
- **Enable `-strict-force-gate` for LJ.** At n=1.0: 67% dual convergence, 2.1 pt loss, 52× force quality improvement. At n=2.0: 10% dual convergence, 2.0 pt loss, 73× force quality improvement.
- **For pairwise potentials**, strict convergence is not just hard to achieve—it is *the wrong criterion*. Ghost eigenvalues at $|\lambda| \sim 10^{-5}$ eV/Å^2 carry no physical significance.

For future work on potentials with flat directions (force fields, classical pair potentials, coarse-grained models), the relaxed criterion should be used from the start.

A Lennard-Jones Potential

The LJ potential is a pairwise sum:

$$V = \sum_{i<j} 4\,\varepsilon_{ij} \left[\left(\frac{\sigma_{ij}}{r_{ij}}\right)^{12} - \left(\frac{\sigma_{ij}}{r_{ij}}\right)^{6} \right], \tag{1}$$

with analytical gradient and Hessian:

$$\frac{\partial V}{\partial r_{ij}} = \frac{4\varepsilon_{ij}}{r_{ij}} \left[-12 \left(\frac{\sigma_{ij}}{r_{ij}}\right)^{12} + 6 \left(\frac{\sigma_{ij}}{r_{ij}}\right)^{6} \right], \tag{2}$$

$$\frac{\partial^2 V}{\partial r_{ij}^2} = \frac{4\varepsilon_{ij}}{r_{ij}^2} \left[156 \left(\frac{\sigma_{ij}}{r_{ij}}\right)^{12} - 42 \left(\frac{\sigma_{ij}}{r_{ij}}\right)^{6} \right]. \tag{3}$$

The Cartesian Hessian off-diagonal 3×3 blocks are:

$$H_{ia,jb} = -\left(\frac{\partial^2 V}{\partial r^2} - \frac{1}{r}\frac{\partial V}{\partial r} \right) \hat{r}_a \hat{r}_b \; - \; \frac{1}{r}\frac{\partial V}{\partial r}\, \delta_{ab}, \quad i \neq j, \tag{4}$$

with diagonal blocks $H_{ii} = -\sum_{j\neq i} H_{ij}$.

UFF parameters and σ-scaling. Per-atom σ_i, ε_i from UFF (Rappé et al., *J. Am. Chem. Soc.* 1992), with Lorentz–Berthelot mixing: $\sigma_{ij} = (\sigma_i + \sigma_j)/2$, $\varepsilon_{ij} = \sqrt{\varepsilon_i \varepsilon_j}$. Raw UFF $\sigma \in [2.6, 3.7]$ Å ($r_{\mathrm{eq}} = 2^{1/6}\sigma \approx 2.9$–$4.2$ Å). We scale by 1/3 (`lj_sigma_scale = 0.3333`) to shrink equilibria to ∼1.0–1.3 Å, matching covalent bond lengths. 1 kcal/mol = 0.0434 eV.

What LJ lacks. V depends only on distances r_{ij}, not angles or dihedrals. Directions changing angles without changing any r_{ij} have *exactly zero* Hessian curvature—the origin of ghost modes (§4).

B The v12b Optimizer

The optimizer is a Newton–Raphson minimizer in Cartesian coordinates, assembled incrementally across v1–v12b.

B.1 RFO step builder (v10b)

The raw Hessian is mass-weighted and projected to the $(3N-k)$-dimensional vibrational subspace (Section B.4). The reduced Hessian is diagonalized:

$$\mathbf{H}_{\mathrm{red}} = \mathbf{Q}_{\mathrm{vib}}^{\top} \mathbf{M}^{-1/2} \mathbf{H} \mathbf{M}^{-1/2} \mathbf{Q}_{\mathrm{vib}}, \quad \mathbf{H}_{\mathrm{red}}\, \boldsymbol{v}_i = \lambda_i\, \boldsymbol{v}_i. \tag{5}$$

The step uses the RFO augmented Hessian (Schlegel 2011):

$$\begin{pmatrix} \mathbf{H}_{\mathrm{red}} & \boldsymbol{g} \\ \boldsymbol{g}^{\top} & 0 \end{pmatrix} \begin{pmatrix} \boldsymbol{s} \\ 1 \end{pmatrix} = \mu \begin{pmatrix} \boldsymbol{s} \\ 1 \end{pmatrix}, \tag{6}$$

with μ found by Newton's method on the secular equation $\sum_i c_i^2/(\lambda_i - \mu) + \mu = 0$. Step coefficients: $\Delta x_i = -c_i/(\lambda_i - \mu)$. For minimization, $\mu < \lambda_{\min} \Rightarrow$ guaranteed descent. RFO automatically interpolates between steepest descent and Newton—no hyperparameters.

B.2 Trust region and displacement capping

The step is capped per-atom: $\Delta\boldsymbol{x} \leftarrow \Delta\boldsymbol{x} \cdot R/\max_i\|\Delta\boldsymbol{x}_i\|$ when $\max_i\|\Delta\boldsymbol{x}_i\| > R$. Trust radius updated via $\rho = \Delta E_{\text{actual}}/\Delta E_{\text{predicted}}$:

$$R \leftarrow \begin{cases} \min(1.5\,R,\ \text{MAD}) & \rho > 0.75 \text{ and at boundary} \\ \max(0.25\,\|\Delta\boldsymbol{x}\|_{\max},\ R_{\text{floor}}) & \rho < 0.25 \\ R & \text{otherwise} \end{cases} \tag{7}$$

$R_{\text{floor}} = 0.01\,\text{Å}$ prevents collapse. Steps violating $d_{\min} = 0.5\,\text{Å}$ are rejected.

B.3 Polynomial line search (v10c)

After each accepted step, a cubic $p(t)$ is fit on $[0, 1]$ using energy and directional derivatives at old/new positions. If an interior minimum $t^* \in (0.1, 0.9)$ with $p''(t^*) > 0$ exists and yields lower energy, the step is refined to $t^*\,\Delta\boldsymbol{x}$.

B.4 Eckart projection

TR modes are removed *by construction*, not threshold filtering. An Eckart basis $\mathbf{B} \in \mathbb{R}^{3N\times k}$ ($k = 5$ or 6) is built from mass-weighted translations and rotations. The complementary $\mathbf{Q}_{\text{vib}}$ gives the full-rank reduced Hessian $\mathbf{H}_{\text{red}} = \mathbf{Q}_{\text{vib}}^\top \mathbf{H}_{\text{mw}} \mathbf{Q}_{\text{vib}} \in \mathbb{R}^{(3N-k)\times(3N-k)}$—every eigenvalue is a genuine vibrational frequency. The gradient is also projected (`-project-gradient-and-v`).

B.5 Kick mechanisms (v12/v12b)

Three perturbation mechanisms escape saddle points when $n_{\text{neg}} > 0$. A **mode tracker** maintains eigenvector continuity across steps.

1. Oscillation kick (v12). Fires when energy oscillation is detected at the trust-radius floor for ≥ 3 consecutive checks. Perturbs along the longest-stuck negative eigenvector: $\boldsymbol{x} \leftarrow \boldsymbol{x} + s \cdot \delta \cdot \boldsymbol{v}_{\text{neg}}$, $s = \text{sign}(\boldsymbol{g} \cdot \boldsymbol{v}_{\text{neg}})$. 50-step cooldown.

2. Blind-mode kick with probe (v12b). Fires when a negative mode persists ≥ 100 steps with gradient overlap < 0.1 and force norm $< 0.1\,\text{eV/Å}$. Probes both $\pm\boldsymbol{v}$ at $1, 2, 5, 10, 20 \times R_{\text{floor}}$; accepts first point where n_{neg} decreases.

3. Late escape (v12b). After step 15k, if $n_{\text{neg}} > 0$: displacement $\alpha = 0.1\,\text{Å}$ along longest-stuck mode in both directions. Accepted only if n_{neg} decreases. 500-step cooldown.

Adaptive kick scaling (v12b). $\delta = \min(C\sqrt{|\lambda_{\text{neg}}|},\ 10\,R_{\text{floor}})$, $\delta \geq R_{\text{floor}}$, with $C = 0.1$.

C Starting Geometries and Noise Model

Starting geometry = midpoint: $\boldsymbol{x}_0 = \frac{1}{2}(\boldsymbol{x}_R + \boldsymbol{x}_{\text{TS}})$. Gaussian noise: $\boldsymbol{x}_{\text{start}} = \boldsymbol{x}_0 + \boldsymbol{\eta}$, $\eta_{ia} \sim \mathcal{N}(0, \sigma^2)$, with $\sigma \in \{0.5, 1.0, 1.5, 2.0\}$ Å. Seed per sample: `noise_seed` + `sample_index` (base seed = 42).

D Full Parameter Table

Parameter	Value	Purpose
Calculator	LJ (UFF, $\sigma/3$)	PES evaluation
Force convergence ε	10^{-4} eV/Å	mean per-atom $\bar{F}$
Relaxed eval. threshold τ	0.01	$= \sqrt{\varepsilon}$ (relaxed only)
Max atom displacement (MAD)	1.3 Å	initial trust radius
Trust radius floor R_{floor}	0.01 Å	prevents TR collapse
Min interatomic distance	0.5 Å	rejects unphysical steps
Noise seed	42	reproducibility
Late escape after	15 000 steps	last-resort saddle escape
Late escape α	0.1 Å	late-escape displacement
Late escape cooldown	500 steps	limits late-escape frequency
Adaptive kick C	0.1	eigenvalue-scaled kick magnitude
Osc-kick patience	3	consecutive oscillation detections
Osc-kick cooldown	50 steps	limits osc-kick frequency
Blind-kick patience	100 steps	mode persistence before kick
Blind-kick overlap threshold	0.1	defines "blind" (low overlap)
Blind-kick force threshold	0.1 eV/Å	kick only when forces are small
Blind-kick probe factors	$1, 2, 5, 10, 20$	multiples of R_{floor}

E Cascade Tables

Table 8: Cascade: converged at each eigenvalue threshold (full v12b 50k and baselines).

Config	0.0	0.0001	0.0005	0.001	0.002	0.005	0.01
n = 0.5 Å							
20k baseline	257	264	266	269	271	273	273
50k baseline	258	265	267	270	272	273	273
full v12b 50k	265	271	271	272	273	273	273
n = 1.0 Å							
20k baseline	168	252	261	264	267	279	279
50k baseline	180	265	271	274	276	278	279
full v12b 50k	185	272	274	278	279	279	279
n = 1.5 Å							
20k baseline	70	218	249	259	268	280	281
50k baseline	92	254	276	278	281	282	282
full v12b 50k	96	265	279	282	282	282	282
n = 2.0 Å							
20k baseline	13	196	260	272	277	282	284
50k baseline	17	209	269	277	281	283	285
full v12b 50k	19	210	275	282	285	285	285

The knee is always at $0 \to 0.0001$: this is where the majority of ghost modes live (+191 samples at n=2.0).

F Step Distributions and Computational Cost

G Gain Decomposition

Kicks contribute less on LJ at every noise level, because the dominant failure (ghost modes) cannot be resolved by perturbation.

Table 9: Step and wall-time distributions (non-error samples).

Config	Min	Max	Mean	Median	P90	Wall mean	Wall med
n = 0.5 Å							
20k baseline	2	15 080	1 276	516	3 224	5.3 s	1.4 s
50k baseline	2	31 986	1 465	522	3 457	7.5 s	1.4 s
full v12b 50k	2	49 680	1 602	375	2 909	6.7 s	1.1 s
n = 1.0 Å							
20k baseline	11	18 737	3 595	2 230	9 345	14.3 s	7.3 s
50k baseline	11	48 278	5 009	2 309	12 557	19.2 s	7.7 s
full v12b 50k	12	50 000	4 958	2 086	12 776	17.7 s	7.2 s
n = 1.5 Å							
20k baseline	1	20 000	7 281	4 062	20 000	21.3 s	12.1 s
50k baseline	1	50 000	10 398	4 062	35 690	30.3 s	12.2 s
full v12b 50k	1	50 000	6 067	3 169	15 734	21.4 s	10.9 s
n = 2.0 Å							
20k baseline	1	20 000	3 494	813	13 888	10.0 s	2.3 s
50k baseline	1	50 000	4 968	813	13 888	14.0 s	2.2 s
full v12b 50k	1	50 000	3 219	817	10 545	11.2 s	2.9 s

Table 10: Median steps and wall time: LJ vs. DFTB0 (full v12b 50k).

	Median steps		Mean wall		Speedup	
Noise	LJ	DFTB0	LJ	DFTB0	Steps	Wall
0.5 Å	375	2 613	6.7 s	74 s	7×	11×
1.0 Å	2 086	5 435	17.7 s	165 s	2.6×	9×
1.5 Å	3 169	7 755	21.4 s	238 s	2.4×	11×
2.0 Å	817	∼11k	11.2 s	368 s	14×	33×

H Failure Eigenvalue Spectra

H.1 Error counts

Every combo produces 2–14 errors from a pre-existing `out_new` `UnboundLocalError` bug (code bug, not optimizer failure).

Config	n=0.5	n=1.0	n=1.5	n=2.0
All configs (identical)	14	8	5	2

H.2 n=0.5: mild ghost modes

Full v12b 50k: only 2 failures. Sample 35: $\lambda_{\min} = -6.7\times10^{-4}$. Sample 180: $\lambda_{\min} = -1.35\times10^{-3}$. Both pass at threshold 0.002.

H.3 n=1.0: single-mode failures

6 failures, all with exactly 1 negative eigenvalue in $(-0.002, 0)$. All ran 50k steps. Not real saddle points.

H.4 n=1.5: multi-mode cascade

20k baseline: 53 failures with 1–10 neg modes. 50k: 15. Full v12b: only 3 (all pass at threshold 0.001).

Table 11: Gain decomposition (strict): LJ vs. DFTB0 v13.

	LJ v2			DFTB0 v13		
Noise	Δ_{50k}	Δ_{kicks}	Total	Δ_{50k}	Δ_{kicks}	Total
0.5	+1 (+0.3)	+7 (+2.4)	+8	+3 (+1.0)	+6 (+2.1)	+9
1.0	+12 (+4.2)	+5 (+1.7)	+17	+12 (+4.2)	+14 (+4.9)	+26
1.5	+22 (+7.7)	+4 (+1.4)	+26	+34 (+11.8)	+19 (+6.6)	+53
2.0	+4 (+1.4)	+2 (+0.7)	+6	+50 (+17.4)	+35 (+12.2)	+85

H.5 n=2.0: ghost mode dominance

Full v12b 50k autopsy: ghost modes 191/287 (67%), drifting 38 (13%), oscillating 33 (12%), strict converged 19 (7%). Only 1 non-converged failure (sample 178, 2 neg modes, 50k exhausted).

I Ablation Results at n=2.0

Table 12: Ablations at n=2.0 Å. Converged includes relaxed-accept.

Configuration	Strict	Converged	Non-conv	Med. steps	Wall
20k baseline (no kicks)	13 (4.5%)	263	22	813	10 s
50k baseline (no kicks)	17 (5.9%)	275	10	813	14 s
Full v12b, 20k	16 (5.6%)	276	9	817	10 s
Full v12b, 50k	19 (6.6%)	284	1	817	11 s
Adapt+probe (no late)	20 (7.0%)	284	1	817	11 s
Probe+late (no adapt)	**21** (7.3%)	**285**	**0**	866	11 s

Key findings: (1) step count matters more than kicks, (2) late escape unnecessary at 50k, (3) adaptive scaling unnecessary on smooth LJ, (4) all 50k configs give 284–285/287 under relaxed—differences are ghost-mode noise, not meaningful.

J Hardest Samples

LJ hardest			DFTB0 hardest		
Sample	Conv	Rate	Sample	Conv	Rate
106	4/15	27%	136	6/15	40%
35	6/15	40%	150	6/15	40%
178	8/15	53%	209	7/15	47%
71	9/15	60%	212	8/15	53%
67	10/15	67%	51	9/15	60%

Zero overlap. Difficulty is PES-specific, not geometry-specific.

K Theoretical Discussion

K.1 Validation of the $\varepsilon^{1/2}$ theorem

The $\varepsilon^{1/2}$ theorem (Boumal et al.) predicts $\lambda_{\min} \geq -C\sqrt{\varepsilon}$ for RFO-type methods. With $\varepsilon = 10^{-4}$: $\lambda_{\min} \geq -0.01$.

LJ provides striking validation: at n=2.0, relaxing to -0.01 rescues 285/287 (99.3%). At n=1.0, all 6 failures have eigenvalues in $(-0.002, 0)$. The cascade knee at 10^{-4}–10^{-3} confirms the theorem's bound is conservative.

On DFTB0 the theorem is invisible (strict $\approx$ relaxed). LJ exposes it by providing genuinely flat directions.

K.2 Pairwise rank deficiency

For $V = \sum_{i<j} V(r_{ij})$, pure angular deformations (bond rotations about the axis connecting terminal atoms) change no pairwise distance to first order. These zero-eigenvalue directions are *intrinsic to the potential.* At equilibrium, coupling through the pairwise network may give quasi-angular modes positive curvature, but noise displaces the geometry into flat regions, increasing the number of near-zero eigenvalues.

K.3 Cascade gap scaling

Noise σ	LJ gap (pts)	DFTB0 gap	Ratio
0.5 Å	2.8	$\sim$0	—
1.0 Å	32.7	$\sim$0	$\sim\infty$
1.5 Å	64.9	$\sim$0	$\sim\infty$
2.0 Å	92.7	$\sim$0	$\sim\infty$

Gap scales roughly as σ^2, consistent with the number of flat directions growing with displacement magnitude.

L Known Issues

1. `out_new` **bug:** 5–14 samples/combo crash with `UnboundLocalError` in the trust-radius retry loop. Code bug, not optimizer limitation.
2. **Seed consistency:** Different seeds produce identical results at n=2.0 full v12b 50k (strict=19, relaxed=285, same single failure sample 178).
3. **Hessian precision:** LJ Hessians computed in float64 to avoid `linalg.eigh` failures from wide eigenvalue spectrum.